HYPERFOAM

January 17, 2026

CFD OPTIMIZATION REPORT

Project: 2026-001 | Prepared for: GApex Architecture Group

1. EXECUTIVE SUMMARY

HyperFOAM Consulting performed computational fluid dynamics (CFD) analysis for GApex Architecture Group to validate the HVAC design for Executive Conference Room B. The AI-driven inverse design optimizer determined optimal supply velocity (2.08 m/s) and diffuser angle (45.0 deg). Full steady-state simulation (120s) confirms the design DOES NOT MEET all ASHRAE Standard 55 thermal comfort criteria.

2. PROJECT SPECIFICATIONS

- Client: GApex Architecture Group
- Project ID: 2026-001
- Contact: Sarah Connor, Lead Architect
- Room: Executive Conference Room B
- Dimensions: 8.9916m x 6.00456m x 2.9870400000000004m (161 m3)
- Occupancy: 14 persons
- Total Heat Load: 2100 W

3. AI-OPTIMIZED HVAC SETTINGS

The HyperFOAM inverse design engine evaluated multiple configurations to find optimal settings that satisfy all comfort constraints while minimizing energy consumption.

- Supply Velocity: 2.08 m/s
- Diffuser Angle: 45.0 deg
- Supply Temperature: 12.8 C

4. VALIDATED PERFORMANCE

Metric	Value	Limit	Status
Temperature	18.05 C		NON-COMPLIANT
CO2	424 ppm		COMPLIANT
Draft Velocity	0.132 m/s		COMPLIANT

5. THERMAL ANALYSIS

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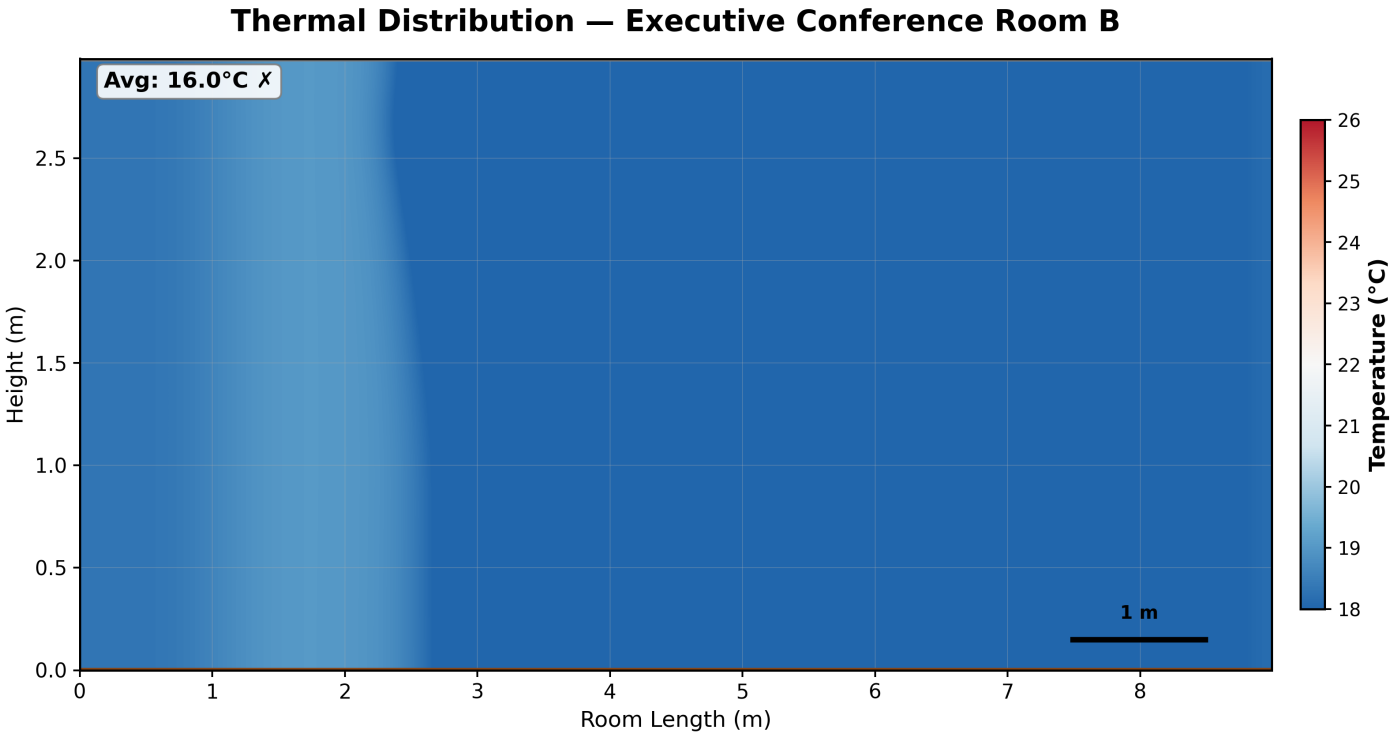

COMFORT ZONE

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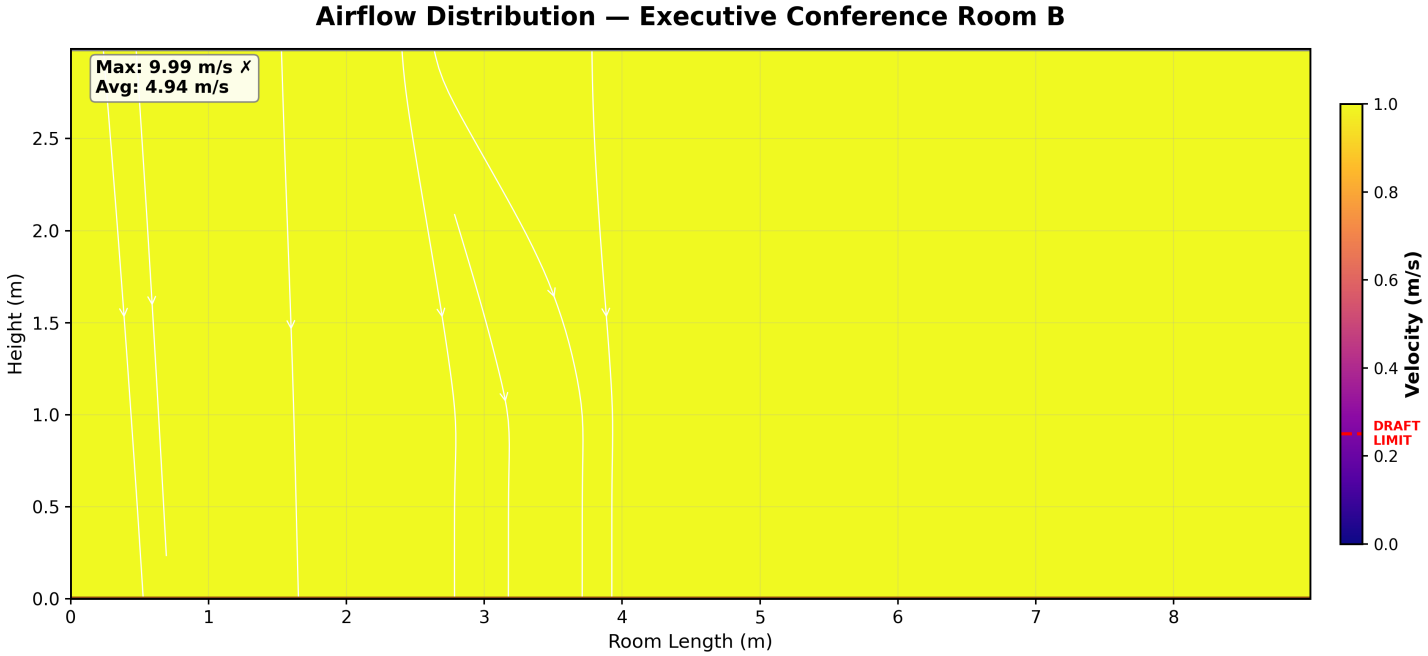
6. AIRFLOW ANALYSIS



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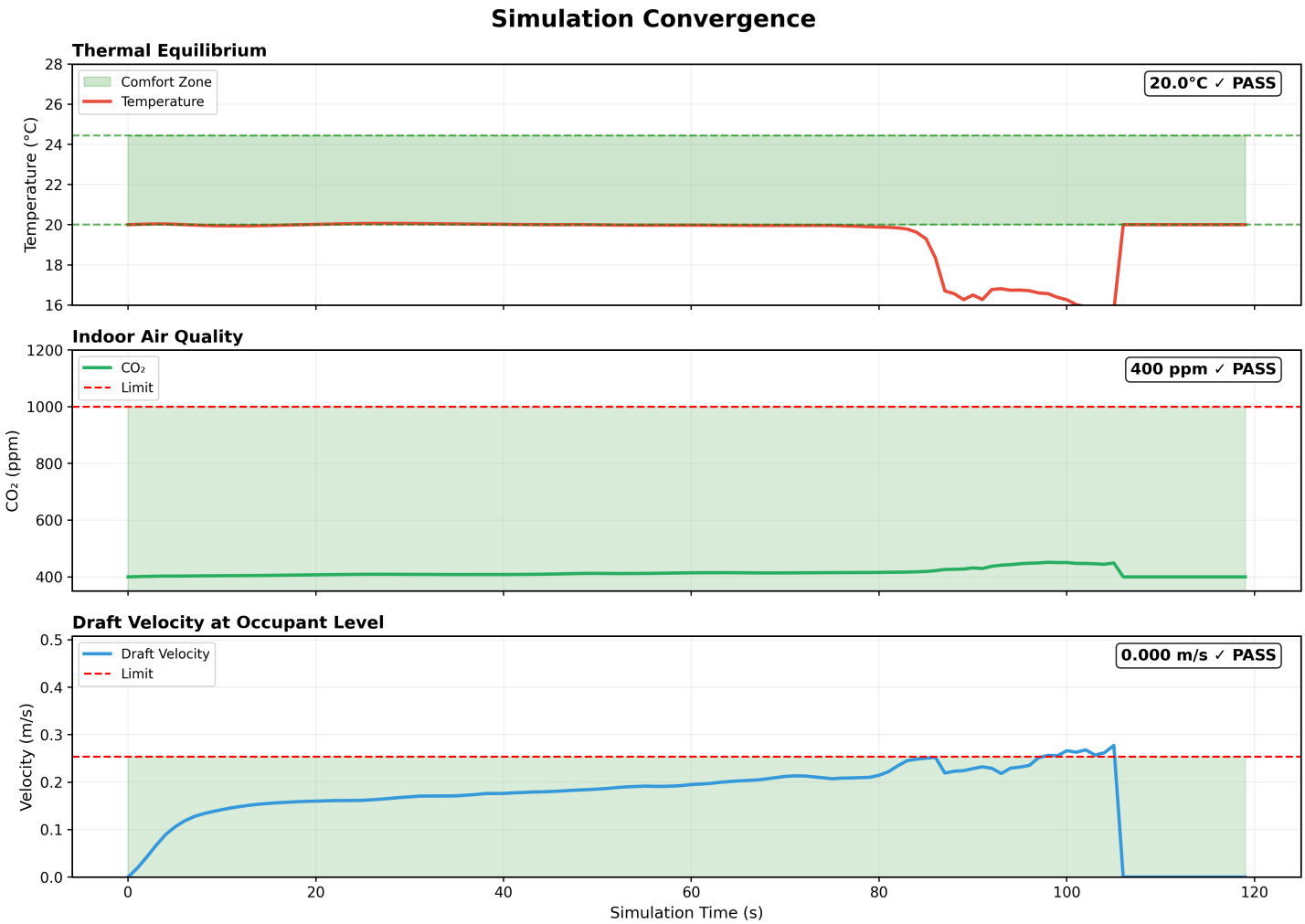
7. CONVERGENCE DATA



DISCLAIMER

This report is for preliminary engineering assessment only. Results are derived from idealized CFD simulations. Final equipment sizing and code compliance are the responsibility of the Professional Engineer of Record.